

Lecture 6: Some techniques to go around the sign problem

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Abstract

In this lecture we revisit and sharpen the distinction between sign quenching and phase quenching in lattice QCD at finite baryon density, reviewing the reweighting formulae and the origin of the sign/phase problem in the thermodynamic limit. We then discuss two conventional strategies for circumventing the sign problem: the Taylor expansion of thermodynamic observables about $\mu/T = 0$ and the analytic continuation from imaginary chemical potential. Finally, we introduce *Lefschetz thimbles* as an alternative approach to the complex-action problem, showing how complexifying the field variables and integrating along steepest-descent manifolds removes the oscillatory phase from the path integral.

1 Recap: Sign vs. Phase Quenching

We begin by re-emphasising that **sign quenching** and **phase quenching** are conceptually distinct procedures, even though both are responses to the complex-action problem that arises in lattice QCD at finite quark chemical potential μ . The complex-action problem is the statement that for $\mu \neq 0$ the fermion determinant $\det M(U, \mu)$ is in general complex, so the integrand $e^{-S_q[U]} \det M(U, \mu)$ cannot be interpreted as a probability weight for importance sampling. Phase quenching replaces $\det M$ by its modulus $|\det M|$ (a continuous positive quantity), while sign quenching replaces it by $\text{Re} \det M$, which is further restricted to positive values by the restriction $|\text{Re} \det M|$ and one keeps only the residual sign factor ± 1 . The two quenching methods therefore give rise to genuinely different probability distributions to sample.

1.1 Phase quenching

The full and phase-quenched partition functions are defined respectively as

$$Z_{\text{full}} = \int \mathcal{D}U \, e^{-S_q[U]} \det M(U, \mu), \quad (1)$$

$$Z_{\text{pq}} = \int \mathcal{D}U \, e^{-S_q[U]} |\det M(U, \mu)|. \quad (2)$$

Writing $\det M = |\det M| e^{i\phi(U,\mu)}$, the expectation value of an observable $\mathcal{O}$ in the full theory can be expressed as a ratio of phase-quenched averages,

$$\langle \mathcal{O} \rangle_{\text{full}} = \frac{\langle e^{i\phi} \mathcal{O} \rangle_{\text{pq}}}{\langle e^{i\phi} \rangle_{\text{pq}}} \approx \frac{\frac{1}{N} \sum_{i=1}^N e^{i\phi(U_i)} \mathcal{O}(U_i)}{\frac{1}{N} \sum_{i=1}^N e^{i\phi(U_i)}}, \quad (3)$$

where the Monte Carlo average on the right is taken with the (positive) phase-quenched weight $|\det M| e^{-S_q}$. The factor $e^{i\phi(U_i)}$ is evaluated as a post-hoc reweighting on each gauge configuration U_i .

What is $\langle e^{i\phi} \rangle_{\text{pq}}$? This average, sometimes called the *average phase factor*, controls the severity of the sign problem. Directly from the definitions above one has

$$\langle e^{i\phi} \rangle_{\text{pq}} = \frac{1}{Z_{\text{pq}}} \int \mathcal{D}U e^{-S_q[U]} |\det M| e^{i\phi} = \frac{Z_{\text{full}}}{Z_{\text{pq}}}. \quad (4)$$

Recognising both Z_{full} and Z_{pq} as exponentials of extensive free energies, $Z = e^{-f\Omega}$ where $\Omega = V/T$ is the four-volume, one obtains

$$\langle e^{i\phi} \rangle_{\text{pq}} = e^{-\Delta f \Omega}, \quad \Delta f \equiv f_{\text{full}} - f_{\text{pq}} \geq 0. \quad (5)$$

Why $\Delta f \geq 0$? By the triangle inequality applied to the path integral,

$$Z_{\text{full}} = \left| \int \mathcal{D}U e^{-S_q[U]} |\det M| e^{i\phi} \right| \leq \int \mathcal{D}U e^{-S_q[U]} |\det M| = Z_{\text{pq}}. \quad (6)$$

Together with $Z = e^{-f\Omega}$ this gives $e^{-f_{\text{full}}\Omega} \leq e^{-f_{\text{pq}}\Omega}$, which is equivalent to $f_{\text{full}} \geq f_{\text{pq}}$, i.e. $\Delta f \geq 0$.

The sign problem in the thermodynamic limit. Equation (5) shows that as $\Omega \rightarrow \infty$,

$$\langle e^{i\phi} \rangle_{\text{pq}} = e^{-\Delta f \Omega} \rightarrow 0, \quad (7)$$

unless $\Delta f \rightarrow 0$, i.e. unless the full and phase-quenched theories happen to be thermodynamically equivalent. This exponential suppression of the denominator in Eq. (3) is the precise statement of the sign problem in the phase-quenching approach.

1.2 Sign quenching

In sign quenching one starts from the observation that physical observables such as Z_{full} must be real (as argued above). Since the integration measure is invariant under $U \rightarrow U^*$ and the action satisfies $S_q[U^*] = S_q[U]$, the imaginary part of the integrand cancels pairwise:

$$Z_{\text{full}} = \int \mathcal{D}U \det M(U, \mu) e^{-S_q[U]} = \int \mathcal{D}U \text{Re}[\det M(U, \mu)] e^{-S_q[U]}. \quad (8)$$

However, $\text{Re}[\det M]$ is *not* guaranteed to be non-negative on every configuration – it merely averages to a positive number – so it still cannot serve directly as a probability

The phase $\phi(U, \mu)$ is called the *complex phase* of the fermion determinant. For $\mu = 0$ the γ_5 -Hermiticity of the Dirac operator forces $\det M$ to be real (in fact non-negative for two mass-degenerate flavours), so the phase vanishes and there is no sign/phase problem.

In your exercise you will verify that the relative statistical error on $\langle \text{sign} \rangle$ in a Monte Carlo estimate (standard deviation divided by mean) scales as $e^{\Delta f \Omega} / \sqrt{N}$: the *signal-to-noise ratio* is exponentially suppressed in the volume. Recovering a fixed precision therefore demands $N \sim e^{2\Delta f \Omega}$ samples – the hallmark of the sign problem.

weight. One therefore defines the *sign-quenched* partition function by taking the absolute value pointwise,

$$Z_{\text{sq}} = \int \mathcal{D}U \, |\text{Redet } M(U, \mu)| e^{-S_q[U]}, \quad (9)$$

and uses the decomposition

$$\text{Redet } M = \text{sign}(U, \mu) |\text{Redet } M|, \quad \text{sign}(U, \mu) \in \{+1, -1\}, \quad (10)$$

to write the full expectation value as a reweighted sign-quenched average,

$$\langle \mathcal{O} \rangle_{\text{full}} = \frac{\langle \text{sign}(U) \mathcal{O}(U) \rangle_{\text{sq}}}{\langle \text{sign}(U) \rangle_{\text{sq}}}. \quad (11)$$

For observables that are themselves invariant under $\mathcal{O}(U) = \mathcal{O}(U^*)$, this representation involves only real numbers and is well-defined as long as $\langle \text{sign} \rangle_{\text{sq}} \neq 0$. The severity of this condition – in particular the exponential suppression $\langle \text{sign} \rangle_{\text{sq}} \sim e^{-(\Delta f)_{\text{sq}}} \Omega$ – will be examined in the exercises.

Phase vs. sign quenching, in one line. Phase quenching uses $|\det M|$ as the proxy measure and corrects with a continuous phase $e^{i\phi} \in U(1)$; sign quenching uses $|\text{Redet } M|$ and corrects with a discrete sign $\pm 1 \in \mathbb{Z}_2$. Both are ratio reweightings of the same form $\langle \mathcal{O} \rangle_{\text{full}} = \langle w \mathcal{O} \rangle_{\text{proxy}} / \langle w \rangle_{\text{proxy}}$. Moreover, since $|\text{Re det } M| \leq |\det M|$ pointwise, one has $Z_{\text{full}} \leq Z_{\text{sq}} \leq Z_{\text{pq}}$ and therefore

$$\langle \text{sign} \rangle_{\text{sq}} = \frac{Z_{\text{full}}}{Z_{\text{sq}}} \geq \frac{Z_{\text{full}}}{Z_{\text{pq}}} = \langle e^{i\phi} \rangle_{\text{pq}} :$$

the sign-quenched average reweighting factor is *never smaller* than the phase-quenched one, i.e. $(\Delta f)_{\text{sq}} \leq (\Delta f)_{\text{pq}}$.

2 Conventional Approaches at Finite Density

We now survey two established methods for computing observables in lattice QCD at finite baryon chemical potential μ , both of which avoid simulating directly at real μ [1, 2, 3].

2.1 Taylor expansion about $\mu/T = 0$

Notice first that the chemical potential always enters as the dimensionless ratio μ/T . A key exact symmetry of the partition function is

$$Z(\mu/T) = Z(-\mu/T), \quad (12)$$

which follows from the combined charge-conjugation and γ_5 -Hermiticity properties of the lattice action. Two exact relations are at work here. (i) γ_5 -Hermiticity, $\gamma_5 M(U, \mu) \gamma_5 = M(U, -\mu^*)^\dagger$, gives $\det M[U, \mu] = \det M^*[U, -\mu^*]$ on the *same* configuration, hence $Z(\mu) = Z^*(-\mu^*)$. (ii) Charge conjugation, $U_\mu(n) \xrightarrow{C} U_\mu^*(n)$, gives $\det M[U, \mu] = \det M^*[U^*, \mu^*]$, hence $Z(\mu) = Z^*(\mu^*)$, i.e. Z is real for real μ . Combining (i) and (ii) yields $Z(\mu) = Z(-\mu)$, Eq. (12).

Subscript “sq” stands for *sign-quenched*

Since $Z(\mu/T)$ is an even function of μ/T , the pressure $p = (T/V) \ln Z$ admits the Taylor series [4]

$$p = T^4 \sum_{n=0}^{\infty} c_{2n}(T) \left(\frac{\mu}{T}\right)^{2n}, \quad (13)$$

or equivalently, writing $\hat{p} \equiv p/T^4$ and $a_{2n} \equiv c_{2n}(T)$ for brevity,

$$\hat{p}\left(\frac{\mu}{T}\right) = a_0 + a_2\left(\frac{\mu}{T}\right)^2 + a_4\left(\frac{\mu}{T}\right)^4 + a_6\left(\frac{\mu}{T}\right)^6 + \dots, \quad (14)$$

$$\hat{p}\left(\frac{i\mu}{T}\right) = a_0 - a_2\left(\frac{\mu}{T}\right)^2 + a_4\left(\frac{\mu}{T}\right)^4 - a_6\left(\frac{\mu}{T}\right)^6 + \dots. \quad (15)$$

The coefficients $c_{2n}(T)$ are – in principle – obtained from $\mu = 0$ simulations via the derivatives of the pressure,

$$c_{2n}(T) = \frac{1}{(2n)!} \left. \frac{\partial^{2n}(p/T^4)}{\partial(\mu/T)^{2n}} \right|_{\mu=0}. \quad (16)$$

This approach is conceptually clean but has two well-known failure modes:

- (1) The higher-order coefficients $\partial^{2n}(p/T^4)/\partial(\mu/T)^{2n}|_{\mu=0}$ are generalised quark-number susceptibilities (cumulants of the baryon-number distribution), whose evaluation involves increasingly many noisy disconnected diagrams. Their stochastic estimation becomes rapidly harder with n : beyond $n \sim 4$ the signal is typically lost in the statistical noise.
- (2) Even if one could compute all coefficients exactly, the series has a finite radius of convergence set by the nearest singularity in the complex μ/T plane. One cannot reach physically interesting regions of the QCD phase diagram (e.g. a putative critical end-point) by truncating the series at low order.

2.2 Analytic continuation from imaginary μ

A complementary strategy exploits the fact that at purely imaginary chemical potential $\mu = i\mu_I$ with $\mu_I \in \mathbb{R}$, the Dirac operator satisfies $\gamma_5 M(i\mu_I) \gamma_5 = M(i\mu_I)^\dagger$, so $\det M$ remains real (and positive in the cases of practical interest, e.g. for two degenerate flavours where the measure is $|\det M|^2$, or for staggered fermions where the eigenvalue structure guarantees $\det M > 0$) [2, 3]. There is therefore no sign problem and one can perform standard Monte Carlo simulations at $\mu = i\mu_I$. This motivates performing exact simulations at imaginary μ and analytically continuing the results to real μ .

Concretely, one computes the (dimensionless) baryon number density

$$\frac{n_B}{T^3} \left(\frac{\mu}{T}, T\right) = \frac{\partial(p/T^4)}{\partial(\mu/T)} = 2a_2\left(\frac{\mu}{T}\right) + 4a_4\left(\frac{\mu}{T}\right)^3 + 6a_6\left(\frac{\mu}{T}\right)^5 + \dots, \quad (17)$$

which at imaginary chemical potential takes the form

$$\frac{n_B}{T^3} \left(\frac{i\mu_I}{T}, T\right) = \frac{i\mu_I}{T} \left[2a_2 - 4a_4\left(\frac{\mu_I}{T}\right)^2 + 6a_6\left(\frac{\mu_I}{T}\right)^4 - \dots \right], \quad (18)$$

purely imaginary as required by symmetry. The coefficients a_{2n} can then be extracted from a fit to lattice data at several values of μ_I and used to reconstruct $p(\mu/T)$ at real μ , within the radius of convergence.

However, at purely imaginary μ , the partition function gets a non-trivial periodicity (Roberge-Weiss symmetry [5]): $Z(\mu_B + i2\pi T) = Z(\mu_B)$. This means there is no "new" information outside the period defined by $\mu = 2\pi T$ interval.

3 Lefschetz Thimbles

Both approaches above ultimately rely on avoiding the sign problem *indirectly*, by simulating at $\mu = 0$ or imaginary μ . A more direct attack on the complex-action problem is offered by the method of *Lefschetz thimbles* [6, 7]. The idea is geometric: rather than trying to evaluate an oscillatory integrand on the original real domain, we *deform* the integration cycle into a complex manifold on which the imaginary part of the action is constant.

3.1 The oscillation problem

Consider the path integral at finite real μ , where the action is complex:

$$Z(\mu) = \int \mathcal{D}\phi e^{-S_R[\phi, \mu] - iS_I[\phi, \mu]}, \quad S = S_R + iS_I, \quad S_R, S_I \in \mathbb{R}. \quad (19)$$

The fundamental difficulty is that the measure $\mathcal{D}\phi = \prod_{i=1}^N d\phi_i$ involves $N \sim 10^6$ – 10^{12} integrations for a state-of-the-art lattice, and the factor $e^{-iS_I[\phi, \mu]}$ oscillates wildly in this enormous space, leading to catastrophic cancellations. A guiding principle from one-dimensional contour integration is that an oscillatory integral can often be tamed by deforming the contour into the complex plane so that the imaginary part of the exponent is stationary – this is the method of *stationary phase* or *steepest descent*. The Lefschetz thimble construction is the rigorous generalisation of this idea to many-dimensional integrals with a holomorphic action.

Crucial assumption. The whole construction requires the action $S[\phi]$ to be a *holomorphic* function of the complexified fields $\phi \in \mathbb{C}^N$. For lattice scalar field theory with polynomial interactions this is automatic. For fermionic theories one integrates out the fermions first: the resulting *weight* $\det M(U, \mu) e^{-S_q}$ is holomorphic in the complexified link variables, since the determinant is a polynomial in them. Note, however, that the effective *action* $S_{\text{eff}} = S_q - \ln \det M$ has logarithmic branch points at the zeros of the determinant; the flow must therefore be defined with the holomorphic weight, and thimbles can terminate at such zeros.

3.2 Complexification of field variables

By Cauchy's theorem, the integration cycle can be deformed in *complexified* field space $\phi \in \mathbb{R}^N \hookrightarrow \mathbb{C}^N$ without changing the value of the integral, provided no singularities are crossed and the integrand decays at infinity. The question is: does there exist an alternative integration cycle

$$\mathcal{D}\phi = \prod_{i=1}^N d\phi_i \longrightarrow \mathcal{D}\phi_c = \prod_{i=1}^N d\phi_{c,i}, \quad \phi_c = \phi^R + i\phi^I \in \mathbb{C}^N, \quad (20)$$

on which $S_I[\phi_c, \mu]$ is **constant**? If so, the constant phase factors out of the integral,

$$Z(\mu) = e^{-iS_I^{\mathcal{J}}} \int_{\mathcal{J}} \mathcal{D}\phi_c e^{-S_R[\phi_c, \mu]} \implies \text{no oscillations in the integrand}, \quad (21)$$

where $S_I^{\mathcal{J}}$ is the (constant) value of S_I on the cycle $\mathcal{J}$. The answer is **yes**: such cycles exist generically and are called *Lefschetz thimbles* [6].

This is a precise version of the sign problem: the integrand oscillates so rapidly that the Monte Carlo estimate of the integral has an exponentially small signal-to-noise ratio in the volume.

The simplest analogy is the passage from a real line integral $\int_{-\infty}^{\infty} dx f(x)$ to a contour integral $\int_C dz f(z)$ in the complex $z = x + iy$ plane. One looks for a contour C on which the imaginary part of the exponent is constant – the *stationary-phase contour*. Equivalently, one finds a curve $\beta(x, y) = \text{const}$ where $f(z) = \alpha(x, y) + i\beta(x, y)$.

3.3 Construction of the thimbles

The construction proceeds in three steps.

- (1) **Find the critical points.** The thimbles are attached to the saddle points ϕ_σ of the complexified action, defined by

$$\left. \frac{\delta S[\phi_c]}{\delta \phi_c} \right|_{\phi_c=\phi_\sigma} = 0. \quad (22)$$

- (2) **Define the steepest-ascent flow.** The thimble $\mathcal{J}_\sigma$ associated with the saddle ϕ_σ is the union of all gradient-ascent trajectories of S_R starting at ϕ_σ :

$$\frac{d\phi^R}{d\tau} = +\frac{\partial S_R}{\partial \phi^R}, \quad (23)$$

$$\frac{d\phi^I}{d\tau} = +\frac{\partial S_R}{\partial \phi^I}, \quad (24)$$

with boundary condition $\phi(\tau \rightarrow -\infty) = \phi_\sigma$. This means that the thimble starts at the critical points defined above. The *unstable-thimble* $\mathcal{K}_\sigma$ is obtained by reversing the sign on the right-hand side: it consists of the steepest-*descent* trajectories emanating from the saddle, along which S_R decreases.

- (3) **Verify that S_I is constant on $\mathcal{J}_\sigma$.** Because $S(\phi_c) = S_R + iS_I$ is holomorphic, the Cauchy–Riemann equations

$$\frac{\partial S_R}{\partial \phi^R} = \frac{\partial S_I}{\partial \phi^I}, \quad \frac{\partial S_R}{\partial \phi^I} = -\frac{\partial S_I}{\partial \phi^R} \quad (25)$$

hold pointwise. Along the flow defined by Eqs. (23)–(24),

$$\begin{aligned} \frac{dS_I}{d\tau} &= \frac{\partial S_I}{\partial \phi^R} \frac{d\phi^R}{d\tau} + \frac{\partial S_I}{\partial \phi^I} \frac{d\phi^I}{d\tau} \\ &= -\frac{\partial S_R}{\partial \phi^I} \cdot \frac{\partial S_R}{\partial \phi^R} + \frac{\partial S_R}{\partial \phi^R} \cdot \frac{\partial S_R}{\partial \phi^I} = 0, \end{aligned} \quad (26)$$

using (25) in the second line. Hence S_I is **constant along every thimble**, as required.

Complex form of the flow. The flow (23)–(24) in real components is equivalent to a single complex equation in terms of the *Wirtinger derivative* $\partial/\partial \bar{\phi} = \frac{1}{2}(\partial/\partial \phi^R + i\partial/\partial \phi^I)$. Combining the two real equations via $d\phi_c/d\tau = d\phi^R/d\tau + i d\phi^I/d\tau$ and using the Cauchy–Riemann relations to write $\partial S_R/\partial \phi^R + i\partial S_R/\partial \phi^I = \partial S/\partial \phi_c$, one obtains the compact form

$$\frac{d\phi_c}{d\tau} = \overline{\left(\frac{\partial S}{\partial \phi_c} \right)}, \quad \frac{d\bar{\phi}_c}{d\tau} = \frac{\partial S}{\partial \phi_c}. \quad (27)$$

This is the form most often used in the literature, as it manifests the holomorphic structure and is well-suited to numerical integration on $\mathbb{C}^N$.

Sign convention. We take the ascent flow with the + sign so that, by construction, S_R increases along the trajectory. This is the parametrisation in which the thimble is traversed *away* from the saddle as τ increases, and the integrand e^{-S_R} is maximal at ϕ_σ .

3.4 Behaviour of S_R along the thimble

Along the ascent flow (23)–(24), the real part of the action behaves as

$$\begin{aligned} \frac{dS_R}{d\tau} &= \frac{\partial S_R}{\partial \phi^R} \frac{d\phi^R}{d\tau} + \frac{\partial S_R}{\partial \phi^I} \frac{d\phi^I}{d\tau} \\ &= \left(\frac{\partial S_R}{\partial \phi^R} \right)^2 + \left(\frac{\partial S_R}{\partial \phi^I} \right)^2 = \|\nabla S_R\|^2 \geq 0. \end{aligned} \quad (28)$$

Hence S_R *increases monotonically* from its value at the saddle ϕ_σ as one moves along the thimble. The integrand e^{-S_R} is therefore *maximal* at the critical point and decays exponentially away from it, guaranteeing convergence of $\int_{\mathcal{J}_\sigma} \mathcal{D}\phi_c e^{-S_R}$. On the anti-thimble $\mathcal{K}_\sigma$ (steepest descent), $dS_R/d\tau = -\|\nabla S_R\|^2 \leq 0$ instead, so S_R *decreases* away from the saddle and the integral diverges – these cycles are therefore unsuitable as integration domains.

Why this is useful. On the thimble, e^{-S_R} acts as a genuine positive weight peaked at the saddle, formally analogous to a Gaussian centred on ϕ_σ . Importance sampling becomes possible, and the overall phase $e^{-iS_I^\mathcal{T}}$ – a single constant – can be pulled out. The exponentially bad signal-to-noise of the original real-domain integral is converted into a much milder problem.

3.5 Thimble decomposition and the residual sign problem

In general the original real integration cycle $\mathbb{R}^N$ can be decomposed as an integer-coefficient sum over stable thimbles attached to the various saddle points [6, 7, 8],

$$Z(\mu) = \sum_{\sigma} m_{\sigma} e^{-iS_{I,\sigma}} \int_{\mathcal{J}_{\sigma}} \mathcal{D}\phi_c e^{-S_R[\phi_c]}, \quad m_{\sigma} \in \mathbb{Z}, \quad (29)$$

so that

$$\langle \mathcal{O} \rangle_{\text{full}} = \frac{\sum_{\sigma} m_{\sigma} e^{-iS_{I,\sigma}} \int_{\mathcal{J}_{\sigma}} \mathcal{D}\phi_c e^{-S_R[\phi_c]} \mathcal{O}(\phi_c)}{\sum_{\sigma} m_{\sigma} e^{-iS_{I,\sigma}} \int_{\mathcal{J}_{\sigma}} \mathcal{D}\phi_c e^{-S_R[\phi_c]}}. \quad (30)$$

Each individual thimble integral is now *free of oscillations* and the only remaining phase factors are the global constants $e^{-iS_{I,\sigma}}$.

This rewriting does not, however, eliminate the sign problem entirely – two *residual* sources of sign cancellation remain in practice:

- (i) **Inter-thimble interference.** If several thimbles contribute with different phases $S_{I,\sigma}$, the sum can suffer cancellations analogous to the original sign problem – though typically much milder, since the number of contributing thimbles is small compared with the dimension of field space. **See Clarifications related to question in lecture at the end of this section.**
- (ii) **Jacobian of the parametrisation.** **We did not go into this complication in the lecture.** The point is that $\phi \rightarrow \phi_c$ is a change of variable – so there must be an associated Jacobian of the transformation. This can in general lead to a complex phase as well, this is called the residual sign problem.

The integers m_{σ} are intersection numbers in algebraic topology: $m_{\sigma} = \langle \mathbb{R}^N, \mathcal{K}_{\sigma} \rangle$ counts how many times the original domain intersects each anti-thimble. They are notoriously hard to compute in practice and constitute one of the main open problems of the thimble approach.

Determining *which* thimbles contribute and *with what multiplicities* m_σ (topological intersection numbers) remains very hard in practice for generic lattice actions, and is the main reason why the Lefschetz thimble programme has not yet replaced reweighting and analytic continuation as the workhorse for finite-density lattice QCD.

Some clarifications:

There were some very good questions in the lecture, one of which I only partially answered and one other question whose answer was only partially correct. Here I offer the correct and hopefully satisfactory answers. Also there was an additional point about the Hessian that you will find in the exercise sheet that I completely forgot to mention:

- **Hessian:** At a critical point the gradient ∂S vanishes, so the leading non-trivial information about the action's local shape lives in the *Hessian* $H = \partial^2 S_R$. By holomorphy, H (a real $2n \times 2n$ matrix) has exactly n positive and n negative eigenvalues: the positive-eigenvalue eigenvectors span the tangent space $T_{\phi_\sigma}(\mathcal{J}_\sigma)$ – the n directions of steepest ascent of S_R *into* the thimble – while the negative ones span the anti-thimble. This is why every thimble construction starts at the saddle: the Hessian gives the only *local* characterisation of which directions the integration should follow, and it is the seed from which the rest of the manifold is grown by the flow.
- **Critical points:** There was a question on the importance of the critical points defined by $\left. \frac{\delta S[\phi_c]}{\delta \phi_c} \right|_{\phi_c = \phi_\sigma} = 0$. This is partially answered in the above point – the critical point is the only place where the gradient vanishes, so the Hessian becomes the leading local information about the action and gives us a well-defined tangent space to start the integration from. But there are deeper reasons too. First, the steepest-descent flow has *fixed points* precisely at critical points, so they are the only places where flow trajectories can asymptotically end – without an endpoint, there is no manifold to build. Second, since every trajectory on $\mathcal{J}_\sigma$ terminates at the same ϕ_σ , continuity pins S_I to the single value $S_I(\phi_\sigma)$ on the entire thimble, which is what allows the constant phase e^{-iS_I} to factor out of the integral. Third, S_R attains its minimum at ϕ_σ and grows monotonically away from it along $\mathcal{J}_\sigma$, so e^{-S_R} is peaked at the saddle and the integral converges automatically.
- **Exactness of the decomposition:** Finally, in Picard-Lefschetz theory the thimbles attached to critical points form a *basis* of the original integration domain, which is what allows the decomposition given by Eq. (30). It is worth stressing that this decomposition is *exact*. This is a direct consequence of Cauchy's theorem applied to a closed holomorphic n -form, not a saddle-point approximation. Restricting to a single thimble $\mathcal{J}_0$, by contrast, is *not* exact: it discards every other contributing saddle, and its accuracy rests on the conjecture that those omitted terms are either non-contributing ($n_\sigma = 0$) or exponentially suppressed by $e^{-[S_R(\phi_\sigma) - S_R(\phi_0)]}$.

The correct answer to the question on what makes the $\langle O \rangle_{\text{full}}$ in Eq. (30) **real**?

One might worry that each individual term $m_\sigma e^{-iS_{I,\sigma}} \int_{\mathcal{J}_\sigma} \mathcal{D}\phi_c e^{-S_R} \mathcal{O}(\phi_c)$ is complex, so how can the sum possibly reproduce a real observable? The resolution is symmetry. For real μ , complex conjugation maps saddles to saddles, $\phi_\sigma \leftrightarrow \bar{\phi}_\sigma$, with $S(\bar{\phi}_\sigma) = S(\phi_\sigma)^*$ and $\mathcal{J}_{\bar{\sigma}} = \overline{\mathcal{J}_\sigma}$. Since the original cycle $\mathcal{C}$ is itself conjugation-invariant, the multiplicities satisfy $m_\sigma = m_{\bar{\sigma}}$, and the contributions pair up:

$$m_\sigma e^{-iS_{I,\sigma}} \int_{\mathcal{J}_\sigma} (\dots) + m_{\bar{\sigma}} e^{+iS_{I,\sigma}} \int_{\mathcal{J}_{\bar{\sigma}}} (\dots) = 2 \operatorname{Re} \left[m_\sigma e^{-iS_{I,\sigma}} \int_{\mathcal{J}_\sigma} (\dots) \right].$$

Saddles that are themselves real (fixed under conjugation) contribute real amounts on their own. The imaginary parts cancel *pairwise and exactly*, just as the imaginary part of $\det M$ cancels under $U \rightarrow U^*$ in the sign-quenching argument. Reality is thus an output of symmetry acting on the saddle set, not a property of the individual thimble integrals.

Lastly, the conjecture I mentioned in the lecture about whether the theory is the same after complexification of the field: The conjecture is that lattice QCD in the continuum limit has only one contributing thimble (the one attached to the classical vacuum). If this is true, all of our problems with complex phases and with having to account for different thimbles go away.

But in general: *the original theory is **not** equivalent to the new theory restricted to a single thimble. What is exact is the decomposition given by Eq. (30).*

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